Ratios and Equivalent Ratio Tables
Answer Key

Grade 6–7 Mathematics

Quick Answers

1. $\frac{3}{4}$
2. $\frac{3}{7}$
3. 42
4. 2, 10
5. 21
6. $\frac{3}{8}$
7. 4, 20
8. —
9. 3, 45
10. 35

Curriculum

Level: Grade 6–7 Mathematics
6.RP.A.2, 6.RP.A.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

2. *If they got 0.75: compared the two parts instead of comparing a part with the whole*
3. *If they got 24: counted only the red boxes instead of all seven boxes*
5. *If they got 80: subtracted the two quantities instead of dividing them*
6. *If they got 0.6: used the other part as the denominator, giving a part-to-part ratio*
10. *If they got 31.5: split the total into two equal halves instead of into 5 + 4 equal shares*

1. Sticker strip: stars to moons, three forms and simplest form.

Step 1 (read the model): count the letters — 6 S stickers and 8 M stickers. Step 2 (write the three forms): 6 to 8, 6:8, $\frac{6}{8}$. All three name the same comparison; only the notation changes. Step 3 (simplify): divide both parts by the common factor 2, giving 3 to 4, or 3:4, or $\boxed{\frac{3}{4}}$.

Note: simplifying does not change the sheet — it says "for every 3 stars there are 4 moons." Accept any of the three forms as the answer, but the fraction must be in simplest form.

2. Same strip: stars to all the stickers.

Step 1 (find the whole): the sheet holds $6 + 8 = 14$ stickers, so this is a part-to-whole comparison, not part-to-part. Step 2 (write it): stars to all is $\frac{6}{14}$. Step 3 (simplify and interpret): $\boxed{\frac{3}{7}}$ — 3 out of every 7 stickers on the sheet is a star. A part-to-whole ratio is the one that may also be read as "that fraction of the group."

3. Tape diagram: red to blue is 4 to 3, each box holds 6 beads.
Step 1 (count boxes): 4 boxes of red and 3 boxes of blue, so the diagram holds $4 + 3 = 7$ equal boxes in all. Step 2 (use the box value): every box holds the same 6 beads, so the bracelet has 7×6 beads. Step 3: $\boxed{42}$ beads — 24 red and 18 blue, which checks, because $24 : 18$ simplifies back to $4 : 3$.

4. Ratio table: 4 red beads for every 6 white beads.
Step 1 (scale down first): from 6 white to 3 white is division by 2, so the red row is divided by 2 as well. Step 2 (read that cell): $\frac{x}{3} = \frac{4}{6}$ gives $6x = 12$, so $x = \boxed{2}$ red beads. Step 3 (scale up to 15): 15 white is 5 times 3 white, so multiply the 2 by 5; as a proportion, $\frac{x}{15} = \frac{4}{6}$ gives $6x = 60$, so $x = \boxed{10}$ red beads. Equivalent ratios always come from multiplying or dividing *both* quantities by the same number.

5. Value of the ratio 84 bottles to 4 minutes.
Step 1 (write the ratio in the asked order): bottles to minutes is 84 to 4, or $\frac{84}{4}$. Step 2 (find the value): the value of a ratio is the quotient of its two numbers, $84 \div 4$. Step 3: $\boxed{21}$ bottles per minute. Subtracting the two numbers would answer nothing about a rate — the value of a ratio is always a quotient.

6. Fiction as a fraction of the whole shelf, from $5 : 3$.
Step 1 (name the parts): $5 : 3$ compares nonfiction with fiction — both are parts, so 5 is not the whole. Step 2 (build the whole): the shelf is $5 + 3 = 8$ equal shares. Step 3 (write the fraction): fiction is 3 of those 8 shares, so the fraction of the shelf that is fiction is $\boxed{\frac{3}{8}}$. That is where the denominator comes from: it is the sum of the two parts, not one of them.

7. Ratio table: 15 tickets cost 12 dollars.
Step 1 (scale down first): from 15 tickets to 5 tickets is division by 3, so divide the cost by 3 as well: $\frac{x}{5} = \frac{12}{15}$ gives $15x = 60$, so $x = \boxed{4}$ dollars. Step 2 (scale up to 25): 25 tickets is 5 times 5 tickets, so multiply that 4-dollar cost by 5. Step 3 (check with the proportion): $\frac{x}{25} = \frac{12}{15}$ gives $15x = 300$, so the block costs $\boxed{20}$ dollars.

8. Compare Mix A (3 scoops in 5 liters) with Mix B (5 scoops in 8 liters).
Step 1 (write both as fractions): Mix A is $\frac{3}{5}$ and Mix B is $\frac{5}{8}$. Step 2 (find each value): $3 \div 5 = 0.6$ scoops per liter and $5 \div 8 = 0.625$ scoops per liter. Common denominators work just as well: $\frac{3}{5} = \frac{24}{40}$ and $\frac{5}{8} = \frac{25}{40}$. Step 3 (compare in the asked order): 0.6 is less than 0.625, so value of Mix A $\boxed{<}$ value of Mix B — Mix B is the stronger mix.

9. Ratio table: 12 lemons for every 8 cups of water.
Step 1 (take the easy step first): divide both quantities by 4 to reach the 2-cup column — 8 cups $\div 4 = 2$ cups and 12 lemons $\div 4 = 3$ lemons, so that cell is $\boxed{3}$. Step 2 (scale up from there): 30 cups is 15 times 2 cups, so multiply the 3 lemons by 15. Step 3 (check with the proportion): $\frac{x}{30} = \frac{12}{8}$ gives $8x = 360$, so the batch needs $\boxed{45}$ lemons.

10. Synthesis: blue to yellow is 5 to 4, and 63 liters of mixed paint are needed.
Step 1 (model the whole): the mix is $5 + 4 = 9$ equal boxes of paint, so the part-to-whole ratio for blue is $\frac{5}{9}$. Step 2 (find one box): the 9 boxes together are 63 liters, so one box is $63 \div 9 = 7$ liters. Step 3 (take the blue boxes): blue is 5 boxes, and $5 \times 7 = 35$, so Ravi needs $\boxed{35\text{ L}}$ of blue paint. Yellow is $4 \times 7 = 28$ L, and $35 + 28 = 63$ L, which checks.